

Product Gap-Closure Backlog

PREPARED BY Energy Hippo implementation team

The UI entry-path review ([entry-path-and-gaps.pdf](#)) found places where EEMSuite supports a capability but has **no UI/data path to create the data**. Per owner decision (2026-06-26), we **close these gaps** rather than defer them — so the dream customer truly exercises everything via the UI. That makes this a **product backlog for the energyhippo/EEMSuite repo** (a sibling repo); this doc specifies each item with the build location the code sweeps found.

Cross-repo: these features are built in C:\Hippo\Git\univerus\energyhippo\EEMSuite (and its Database\EEMSuite), not in this synthetic-customer repo. Once built, the corresponding synthetic data becomes UI-enterable and moves into the materialize phase here.

Effort: **S** ≈ hours–1 day · **M** ≈ a few days · **L** ≈ 1–2+ weeks.

How to use this document

This is the **canonical, standalone reference** for the EEMSuite product features to build/connect so the synthetic customer can exercise everything through the UI. Each item below is self-contained — actionable cold in a future session (including one opened directly in the **energyhippo/EEMSuite** repo). When you pick one up: update its row in the **status tracker**, open the cited files, and implement. Discovered while building the Northlake dream customer; these are genuine product gaps, useful beyond this dataset.

Status tracker

#	Item	Type	Effort	Status
1	Connect ENERGY STAR PM (sandbox)	external/ops	S–M	Not started
2	AP/GL export builder procs (richer ERP formats)	dev (procs)	M–L	Not started
3	GL user-defined-fields view	missed-screen	S	Not started
4	Accruals create/reverse feature	needs-feature	L	Not started
5	Per-meter rate-class/option binding	needs-feature	M	Not started
6	Non-Unique rate classes	needs-feature	S–M	Not started
7	Benchmark definition UI	needs-feature	M	Not started
8	MFR external-email recipients	missed-screen	S	Not started
9	EnergyAI enrollment + connection	needs-feature + external	L	Not started
10	GreenButton (ESPI XML) import	re-enable/build	M–L	Not started

(Detail for each item is in the matching numbered section below. The cited code paths are the build locations the entry-path sweeps found; verify against current code before implementing — the codebase moves.)

Decisions applied

- **ENERGY STAR scores** → connect a **sandbox/real Portfolio Manager account** + run the sync task (B1).
- **AP export formats** → **commission dev** for the richer ERP builder procs (A1).
- **All other gaps** → **build the features** (A2–A8, B2).

1. Connect ENERGY STAR Portfolio Manager (B1) — config/ops, effort S–M

Goal: populate ESResponseFacility scores so Rating History/Comparison, RequestLog, and PerformanceSnapshot's ES score work. **Not product dev** — integration/ops:

1. Obtain an ENERGY STAR PM **sandbox** account + register the test properties (the #/ESPropertySetup data already maps them; PMPropertyID is 0 until linked).
2. Set the CustomerServer (and PM credential) system constants so #/ESNewRatingRequest submits and the **EnergyStar sync task** can call PM.
3. Run EEMSuite.Tasks.EnergyStar (RatingSyncService) to pull score metrics → SaveResponseAsync writes ESResponseFacility. **Unblocks:** Phase 5 ENERGY STAR rating reports. **Note:** scores then reflect PM's model on the synthetic usage — document them as PM-derived, not hand-set.

2. AP/GL export builder procs — richer ERP formats (A1) — effort M (per format), L (all)

Goal: more than the 2 built-in formats are runnable from the AP Approval & Upload screen. **Build (Database repo, CustomDevelopment\Northlake\ pattern):** author the *_UploadFile_Get / *_BillsForApproval_Get builder procs + register each via an ExportSetup/Exports_Get row and a SiEAppObjects EX object. Layouts are already specified in [phase3](#) (from the AP code sweep):

- n1_ap_gludf — fixed-width header+detail with GL-UDF segments (Sacramento style)
- n1_ap_voucher — PeopleSoft AP voucher (C/H/L/D records)
- n1_gl_journal — PeopleSoft GL journal (H/L, balanced debit/credit)
- n1_sap_fi — SAP FI comma-delimited
- n1_ar_interface — AR interface (tenant rebilling) **Depends on:** GL-UDF data (item 3) for the GL-coded formats. **Unblocks:** Phase 3 "one file of each format"; AccountsPayableDetail depth. **No format-builder UI is built** (out of scope) — formats remain dev-authored; this is the accepted model.

3. GL user-defined-fields view (A3) — effort S (missed-screen)

Goal: enter the 5 GL segments (Fund/Dept/Program/Activity/Object) + values. **Build (DBAdmin WPF):** wire a XAML view to the existing `Models/Companies/GLUserDefinedModel.cs` + `GLAccountUDFValueModel.cs` and the existing API (`GLAccountsDTOController.SaveGLAccountFieldAsync` / `SaveGLAccountFieldValueAsync`). Backend already complete — just the view. **Unblocks:** `rp_GLAccountUDV`; GL-coded export segments (item 2).

4. Accruals create/reverse feature (A2) — effort L (needs-feature)

Goal: create an accrual and reverse it through the UI. **Build (Domain + Data + Web API + frontend):** a service + endpoints + screen to (a) create an accrual bill tagged commodity 2300, (b) call the existing `rp_OpenAccruals@Reverse` to produce the 2301 reversal + `BillAccrualMap` pairing. Today no UI/API/service exists; the proc does. **Unblocks:** `OpenAccruals/accruals/mpaaccruals` (Phase 3 + Phase 4 tenant accruals).

5. Per-meter rate-class / option binding (A4) — effort M (needs-feature)

Goal: bind a meter to a specific rate class **option** (voltage/phase/ tax-exempt/farm). **Build (Rate/TenantRebill UI + model):** add `RateClassId/RateClassOptionId` to the meter ↔ rate assignment (`UtilityRateConfig` → `UtilityMeterRates`) and expose a picker in `TenantRebillView`; ensure the engine honors the bound option (today it applies class defaults). **Unblocks:** Phase 2 schedule #9; `rp_RateClassAssignment` depth.

6. Non-Unique rate classes (A5) — effort S-M (needs-feature)

Goal: create the always-applied "All" class (for tax/min-bill scoping). **Build (Rate Modeler):** allow selecting `RateClassType` on create (today hardcoded to `Unique` in `RateModelerAddItemDrawerViewModel`); verify engine handling of the "All" class. **Unblocks:** cleaner Phase 2 tax/min-bill scoping.

7. Benchmark definition UI (A6) — effort M (needs-feature)

Goal: define benchmarks/EUI targets (today benchmark UI is read-only). **Build (Web):** a create/edit screen over `wh_Benchmark` (+ EUI/target/baseline fields) and a save endpoint (`BenchmarkController` is GET-only today). **Unblocks:** Phase 5 custom benchmarks; `BenchmarkRanking` definitions.

8. MFR external-email recipients (A7) — effort S (missed-screen)

Goal: add ad-hoc external email recipients to an MFR delivery. **Build (Web MFR delivery picker):** expose the already-modeled `Recipient.Email` field in `MFRDelivery` UI (domain supports it; picker only offers users/groups). **Unblocks:** Phase 6 external delivery. (100+ recipients already work via groups.)

9. EnergyAI enrollment + connection (B2) — effort L (needs-feature + external)

Goal: produce EnergyAI_Insights for the report. **Build + connect:** (a) an **enrollment UI** over EnergyAI_Loads/_WeatherSites/_TuningPeriod/_BaselinePeriod (today DB-only, no screen); (b) connect the external EnergyAI service + run EEMSuite.Tasks.EnergyAIClient (upload → download). **Unblocks:** Phase 6 rp_EnergyAIInsights. Lowest priority.

10. GreenButton (ESPI XML) import (A8) — effort M-L (re-enable/build)

Goal: ingest GreenButton interval XML. **Build:** re-enable the legacy GreenButton upload path (tasks marked TargetKind: Skip in workers; read-only API only today) or build a small import UI/task over the existing GreenButtonConverterService. **Unblocks:** Phase 6 GreenButton intake. Optional/low priority.

Suggested build order

1. **GL-UDF view (3, S) + MFR external-email (8, S)** — quick missed-screens.
2. **AP builder procs (2, M) + ENERGY STAR sandbox connect (1, S-M)** — high report value.
3. **Per-meter rate option (5) + non-Unique class (6)** — completes the rate story.
4. **Benchmark UI (7)** — analytics.
5. **Accruals (4, L)** — larger feature.
6. **EnergyAI (9, L) + GreenButton (10)** — lowest priority / optional.

What this does NOT change

Everything classified UI-DIRECT / UI-THEN-RUN / DATA-PATH / SYSTEM-GENERATED in the register is unaffected and proceeds to materialize as planned. This backlog only covers the gap items.